W-C-421f

July 25, 1963

SUPERSEDING
Int. Fed. Spec. W-C-00421e (GSA-FSS)
September 4, 1962 and
Fed. Spec. W-C-421a,
November 5, 1940

FEDERAL SPECIFICATION

CLEANER, VACUUM, (ELECTRIC, PORTABLE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The vacuum cleaners covered by this specification are floor and hand models for commercial, household, and hospital use in picking up dry and, when specified, wet materials. Cleaners use the dry filter separation method except type I, class D, which may separate by either the dry or wet filtration method. Type I cleaners are floor models of the enclosed-filter type, cleaning by suction only (no agitator); and type II are floor models of the exposed-filter type, with or without agitator, and hand models. They are operated by electrical motors ranging from 120-watt input to 7-1/2 horsepower. (See section 6 for intended use.)

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, sizes, and styles.* Portable electric vacuum cleaners in accordance with this specification shall be of the following types, classes, sizes, and styles as specified in the invitation to bid, contract or order (see 6.1):

Type I.—Enclosed filter.

Use.—Commercial and industrial.

Class A.—2000 cu. in. minimum separator receptacle capacity.

Size 1.—7/8 to 1-1/2 horsepower.
Style 1.—1-1/4 to 1-1/2 horsepower.
Style 2.—7/8 to 1-1/2 horsepower.
Size 2.—3 horsepower (min.)
Size 3.—5 horsepower (min.)
Size 4.—7-1/2 horsepower (min.)

Class B.—690 cu. in. minimum separator receptacle capacity.

Size 1.—1/2 to 3/4 horsepower.
Style 1.—3/4 horsepower (min.)
Style 2.—1/2 horsepower (min.)
Size 2.—7/8 to 1-1/2 horsepower.
Style 1.—1-1/4 to 1-1/2 horsepower.
Style 2.—7/8 to 1-1/2 horsepower.

Class C.—360 cu. in. minimum separator receptacle capacity.

Size 1.—9,500 r.p.m. to 17,000 r.p.m.
Style 1.—1/3 horsepower (min.) (9,500 r.p.m.)
Style 2.—1/3 horsepower (min.) (17,000 r.p.m.)

Use.—Household.

Class D.—Household.

Size 1.—400 to 850 watts.
Model 1.—Horizontal cylinder.
Model 2.—Vertical canister.

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Use.—Hospital.

Class G.—Hospital.
Size 1.—3/4 to 1 horsepower.

Type II.—Exposed filter.

Use.—Household.

Class A.—Floor model, suction.
Class B.—Floor model, suction and power-driven agitator.
Size 1.—Light duty.
Size 2.—Heavy duty.
Class C.—Hand model, suction only.
Class D.—Hand model, suction and power-driven agitator.

Use.—Commercial and industrial.

Class F.—Floor model, lightweight.
Size 1.—1/3 horsepower (min.)
Size 2.—3/4 horsepower (min.)
Size 3.—1 horsepower (min.)

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CC-M-636—Motors, Fractional Horsepower (Alternating-Current).
CC-M-641 — Motors; Alternating-Current, Integral-Horsepower.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-591—Boxes; Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock Corner.
PPP-B-636—Box, Fiber.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

JAN-P-100 — Packaging and Packing for Overseas Shipment, General Specification.
MIL-C-104—Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116—Preservation, Methods of.
MIL-C-132—Crates, Open, Wood, Maximum Capacity 2500 Pounds.
MIL-L-10547 — Liners, Case, Waterproof.
MIL-I-11748—Interference Reduction for Electrical and Electronic Equipment.
MIL-P-16298 — Preservation, Packaging, Packing, And Marking of Electric Machines Having Rotating Parts and Associated Repair Parts.
MIL-I-16910 — Interference Measurement Radio, Methods and Limits; 14 Kilocycles to 1000 Megacycles.
MIL-I-26600—Interference Control Requirements Aeronautical Equipment.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Standards Association Inc., Standard:

C73—Outlet Receptacles, Attachment Plug Caps and Appliance Plug.

(Copies may be obtained from the American Standards Association Inc., 70 East Forty-Fifth Street, New York 17, New York.)

Underwriters' Laboratories, Inc. Standard:

No. 73—Motor-Operated Appliances.

(Copies may be obtained from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, N. Y.; 207 East Ohio Street, Chicago 11, Ill.; or 1655 Scott Lane, Santa Clara, Calif.)

3. REQUIREMENTS

3.1 Material. The materials used shall be suitable for the purpose intended and shall be free from defects affecting the appearance or which may affect serviceability of the cleaners.

3.2 Fire and casualty hazards.

3.2.1 Each bidder shall submit to the contracting agency proof that the cleaners he proposes to supply under this specification conform to the requirements of the Underwriters' Laboratories, Inc. Standard No. 73 for Motor-Operated Appliances. The label, or listing with re-examination, of the Underwriters' Laboratories, Inc. may be accepted as evidence that the cleaners conform to this requirement.

3.2.2 In lieu of the label, or listing with re-examination, the bidder may submit independent proof, satisfactory to the contracting agency, that the cleaners conform to the applicable requirements of the published standards including methods of tests of the Underwriters' Laboratories, Inc. Standard No. 73.

3.2.3 Compliance with Underwriters' Laboratories, Inc. requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his cleaners. In event of difference between any of the requirements of this specification and those of Underwriters' Laboratories, Inc., the requirements of this specification shall govern.

3.3 General. Each vacuum cleaner shall be complete, fully lubricated, ready to operate, simple in construction, reliable in operation, and constructed so as to be readily disassembled for inspection, adjustment, or replacement. Each part of the vacuum cleaner shall be interchangeable with the similar part of any vacuum cleaner of the same model and size made by the same manufacturer. Class G hospital use, cleaners shall be quiet and nonobjectionable to use in patient areas.

3.4 Identification marking. Each vacuum cleaner shall be plainly marked with the manufacturer's designation for the type, model, size and style, and either his name or trademark of such known character that the source of manufacture may be readily determined.

3.4.1 *Nameplate data.* Nameplate data for type I, classes A, B, C, G, and type II, class B, size 2, and class F shall be given in terms of amperes taken by the motor at an electrical load point determined to be midway between the electrical loads of the motor when the suction air intake to the cleaner is sealed and when the intake is wide open. Nameplate data for type I, class D, and type II, classes A, B, C, and D shall be given in terms of voltage, wattage, and/or amperage. The markings shall appear either on the motor nameplate or at some other visible place on the vacuum cleaner. In addition to the

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foregoing, alternating current motors, exclusive of universal series motors, shall be marked with rated phase and frequency. Constant speed motors shall give nameplate data in terms of voltage and frequency rating with speed and horsepower shown in accordance with NEMA Standards (latest revision).

3.4.2 *Department of Defense marking.* Each vacuum cleaner and components for the military services shall be marked in accordance with MIL-STD-130.

3.5 **Exposed surfaces.**

3.5.1 *Exterior.* The entire exposed surface of each vacuum cleaner shall be smooth. Ferrous-metal parts except corrosion-resisting metal shall be nickel plated, cadmium plated, chromium plated, or primed and suitably coated in accordance with best commercial practice.

3.5.2 *Interior.* The interior of the tank used for "wet" pickup or "wet" separation shall be corrosion-resistant, detergent- esistant metal of thickness to withstand the duty to be performed or have a corrosion-resistant finish. Tanks of other material than metal may be used when specified (see 6.1). Tanks of other than metal shall have a service-durability equal to a metal tank.

3.6 **Stability.** If the vacuum cleaner (except type II, classes C and D) with all parts in operating position, is placed in any position on a plane inclined at 15 degrees to the horizontal, each of the supports of the vacuum cleaner shall remain in contact with the plane surface, when the dirt container is empty or filled with at least one pound of sand.

3.7 **Dirt separation.**

3.7.1 *"Dry" pickup.* Except for type I, class D household vacuum cleaners, the dirt shall be separated by the commercially designated "dry" method. For type I, class D household vacuum cleaners, the dirt shall be separated either by the "dry" or "wet" method at the option of the contractor.

3.7.2 *"Wet" separation.* Vacuum cleaners employing the "wet" method shall be fitted with a conspicuous warning plate stating that the water container should be filled to the required level before using the vacuum cleaner and immediately emptied and dried after each period of use. The water level shall be indicated in a suitable manner. Type I, class D, model 2 vacuum cleaners using the receptacle as a filter and employing the "wet" method of separation, shall pick up dry materials and filter by water only.

3.7.3 *"Wet" pickup.* When specified (see 6.1), the vacuum cleaner shall be designed and constructed for "wet" pickup, that is, the picking up of liquids. The "wet" pickup cleaner shall be provided with (*a*) automatic cut-off assembly to assure protection against overfilling, corrosion-resistant tank as specified in 3.5, tools and all accessories required to convert the cleaner from "dry" to "wet" pickup; and/or when specified, (*b*) dump valve for draining to obviate necessity for lifting water container.

3.8 **Vacuum producer.** The vacuum producer shall be enclosed, shall be of rugged construction, and shall function without objectionable noise or vibration, for the application intended. Surfaces of impellers and air passages shall be constructed so that dirt adherence to the surface of the air passages will not affect the flow of air.

3.9 **Motor.** Except as specified herein, direct-current motors shall be in accordance with, or equal to, the best commercial practice in effect on date of invitation for bids. Alternating-current motors of fractional horsepower size shall be in accordance with CC-M-636 and those of integral horsepower size in accordance with CC-M-641. All motors shall be suitable for the voltage and current specified and shall have class A, B, or H insulation and shall be of adequate capacity to operate under full rated load conditions in an ambient temperature in accordance with requirements of 4.5 without exceeding allowable temperature rises. The 3, 5, and 7-1/2 horsepower motors shall function satisfactorily on the current and voltage stated in the invitation for bids. Motors up to and not ex-

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ceeding 1-1/2-horsepower full load output, shall be of the universal type, suitable for 115-volt direct current and also 60-cycle alternating current; except that the motor shall be suitable for efficient operation on electrical current with other characteristics when specified in the invitation to bid as the available power. When specified (see 6.1), an explosion-proof thermal overload device shall be furnished.

3.9.1 Unless otherwise specified (see 6.1), the motor shall be the size specified in table I and of the duty hereinafter specified for types I and II. The motor shall be either protected (semi-enclosed) or totally enclosed, at the option of the contractor, except that for type I, class G, the motor shall be totally enclosed. Where air is used from the vacuum producer for cooling purposes, it shall be exhaust air so filtered that no dirt will be circulated through the motor.

3.9.2 *Insulation resistance.* With the motor connected to rated source of power supply, and at the maximum operating temperature, the insulation resistance between parts which carry the current and metallic parts which do not carry the current shall be not less than one megohm.

3.9.3 *Dielectric strength.* The insulation shall not fail (break down) when tested under conditions specified in 4.4.2.

3.9.4 *Armature or rotor.* The armature or rotor of the motor shall be accurately balanced and shall be removable. The armature or rotor shall not have objectionable end play or vibration.

3.9.5 *Armature bearings.* Bearings shall be arranged to prevent the lubricant from leaking out of the bearings and shall be provided with suitable means of luricating or shall be of the sealed type requiring no lubrication. Bearings shall be lubricated as necessary for service operation (see 4.6.2).

3.9.6 *Brushes.* The motor brushes, if any, shall be readily removable and of such hardness and wearing properties as to give the service life specified (see 3.16.7.3 and 3.17.5.2).

3.10 **Radio filter.** Unless otherwise specified (see 6.1), when required to effectually climinate interference with reception by radio receivers of standard broadcast bands, commutator type of motors shall be equipped with a radio filter permanently mounted on the vacuum cleaner.

3.10.1 *Radio interference reduction.* Equipment procured under this specification for military use, or for others when specified (see 6.1), shall be equipped for the reduction of radio interference in accordance with MIL-I-11748, class IIId, over the frequency range of 0.15 to 1000 megacycles, MIL-I-26600 (USAF) or MIL-I-16910 as specified (see 6.1).

3.11 **Motor starter or switch.** Except as provided below in this paragraph, a reliable, adequately insulated switch of the enclosed, quick-make-and-break type shall be secured on the vacuum cleaner for stopping and starting the motor. The rating of the switch shall be not less than the rating of the motor. All motors of 3-horsepower rated output and larger shall have an adequate thermal overload device in the electrical circuit. Where the motor is for operation on direct current and exceeds 1-1/2-horsepower rated output, there shall be an adequate starter of the enclosed magnetic type operated by push buttons or other suitable switch. Sufficient steps of resistance shall be provided, where necessary for gradual acceleration of all direct-current motors up to normal speed. When specified (see 6.1), an explosion-proof thermal overload device shall be furnished.

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TABLE I.—*Requirements for vacuum cleaners*

Type	Class	Size	Style	Motor: Output brake horsepower c/	Motor: Input watts	Motor: Load speed (max.) a/ or d/	Cord - 2-wire conductor: Type	Cord: 115 volts d.c. and Univ. (min.)	Cord: 115 volts 1 ph. (min.)	Cord: 220-230 volts d.c. and Univ. (min.)	Cord: 220-230 volts 1 ph. (min.)	Length (min.)	Suction at end of 25 feet of hose: 7/8 inch orifice Col. "X" Water (min.)	7/8 inch orifice Mercury (min.)	5/8 inch orifice Water (min.)	5/8 inch orifice Mercury (min.)	Length of nozzle on cleaner (min.)	Length of hose (min.)	Total weight without attachments (max.)	Separator receptacle capacity (min.)
						r.p.m.		Amer. Wire Ga.				ft.	in.	in.	in.	in.		ft.	lbs.	cu. in.
I	A	1	1	1¼-1½	-	a/ 8,000	S	12	-	-	-	50	33.5	2.5	45	3.3	-	50	450	2,000
			2	7/8-1½	-	d/ 13,000	SJ	14	-	-	-	50	33.5	2.5	42	3.1	-	10	100	2,000
		2	-	3 (min.)	-	a/ 3,600	S	10	8	14	12	50	37	2.75	53	3.9	-	50	1,200	3,200
		3	-	5 (min.)	-	a/ 3,600	S	6	6	12	10	50	41 b/	3.05 b/	57 b/	4.2 b/	-	100	1,200	3,200
		4	-	7½ (min.)	-	a/ 3,600	S	4	4	12	8	50	47.6 b/	3.5 b/	68 b/	5.0 b/	-	100	1,200	5,000
	B	1	1	3/4 (min.)	-	a/ 9,000	S	14	-	-	-	50	22	1.6	33	2.5	-	37	190	690
			2	1/2 (min.)	-	d/ 17,000	SJ	14	-	-	-	50	22	1.6	33	2.5	-	10	75	690
		2	1	1¼-1½	-	a/ 8,000	S	12	-	-	-	50	33.5	2.5	45	3.3	-	50	270	690
			2	7/8-1½	-	d/ 13,000	SJ	14	-	-	-	50	33.5	2.5	42	3.1	-	10	100	690
	C	1	1	1/3 (min)	-	a/ 9,500	S	16	-	-	-	50	16	1.2	26	1.9	-	25	65	360
			2	1/3 (min)	-	d/ 17,000	SJ	16	-	-	-	50	22	1.6	33	2.5	-	10	65	360
	G	1	-	3/4-1	-	20,000	S	14	-	-	-	50	28	2.1	42	3.1	-	10	100	1,450
II	F	1	-	1/3 (min)	-	d/ 17,000	SJ	16	-	-	-	50	16	1.2	26	1.9	-	10	65	1,200
		2	-	3/4 (min)	-	d/ 13,000	SJ	14	-	-	-	50	22	1.6	25	1.8	-	10	100	1,200
		3	-	1 (min.)	-	d/ 13,000	SJ	14	-	-	-	50	30.0	2.5	42	3.1	-	10	100	1,200

a/ Speed shall be determined with orifice open under conditions in column "X". Also see 3.12 for wire size.
b/ Two-cleaner unit, suction at each end of two 50-foot hose coupled in parallel.
c/ Motor horsepower shall be designated as continuous shaft output under the maximum load on the vacuum cleaner at rated speed and voltage.
d/ Speed shall be determined with end of standard length of hose fully open.

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TABLE I.—*Requirements for vacuum cleaners (cont'd).*

Type	Class	Size	Motor: Output, brake horsepower	Motor: Input, watts	Motor: Load speed (max.) a/	Cord: Types	Cord: 115 volts D. C. and univ. (min.)	Cord: 115 volts 1 ph. (min.)	Cord: 220-230 volts D. C. and 3 ph. (min.)	Cord: 220-230 volts 1 ph. (min.)	Cord: Length (min.)	Suction at end of 7 feet 6 inches of hose, 1-inch orifice: Col.X Water (min.)	Suction at end of 7 feet 6 inches of hose, 1-inch orifice: Mercury (min.)	Suction at end of 7 feet 6 inches of hose, 1-inch orifice: Water (min.)	Suction at end of 7 feet 6 inches of hose, 1-inch orifice: In.	Length of nozzle on cleaner (min.)	Length of hose (min.)	Total weight without attachments (max.)	Separator receptacle, capacity (min.)	Dust bag: Side area, Sq.in.	Dust bag: Opening Sq.in.
					r.p.m.		Amer. Wire gage No.				Ft.	In.	In.	In.	In.	In.	Ft.	Lbs.			
I	D	..	.....	400/550	15,000	SV or SVT	18	..	..	..	20	6	...	..	...	...	7'8"	21	200 cu.in. if enclosed dust bag or 4 qt. receptacle if wet separation is used.		
												Suction-Fig. 1 at the box, 1-inch orifice									
II	A	..	.....	275/400	14000	SV or SVT	18	...	...	...	20	7	...	...	...	11	7'8"	17	550	33	
	B	1	.....	275/400	14000	SV or SVT	18	...	...	...	20	5-1/2	...	..	...	11	7'8"	27	550	33	
		2	1/4 (min.)	.....	14000	SV or SVT	16	...	...	...	35	7	...	..	...	11	7'8"	30	550	33	
												Suction-Fig. 1 at the box, 1/2-inch orifice									
	C	..	.....	120/200	14000	SV or SVT	18	...	...	...	20	6	...	..	...	5	...	6	120	..	
	D	..	.....	150/400	14000	SV or SVT	18	...	...	...	20	6	...	..	...	5	...	7	120	..	

a/ Speed shall be determined with orifice open under conditions in column X.

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TABLE II.—*Wire sizes for 3-conductor cords*

Type	Class	Size	Style	Wire gage for 3-conductor cables	
				With 3-current-carrying conductors (minimum)	With 2-current carrying conductors and 1 conductor used as ground only (minimum)
I	A	1	2	14	16
I	B	2	2	14	16
II	F	2	-	14	16
II	F	3	-	14	16
I	B	1	2	14	16
I	C	1	2	16	18
I	G	1	-	14	16

3.12 Cord. A conducting cord(s) in accordance with tables I and II shall be connected to the motor leads at one end, and to a suitable molded attachment plug at the other end. Where 3-conductor electrical cords are furnished and one of the wires is used as a grounding conductor only, cord gages can be as specified in table II. Plugs for type S, SJ, or SV cord shall be of rubber. Plugs for SVT and SRT cord shall be of thermoplastic material. The cord shall be adequately reinforced at the cleaner end. The length of the cord is the distance from the exposed surface when it leaves the cleaner to the prong surface of the plug. Wire sizes of tables I and II are minimum. The wire gage shall be large enough for the machine to operate efficiently with the length cord specified.

3.12.1 *Grounding conductor and cord.* For all Department of Defense purchases, the cord shall include a grounding conductor which shall be connected to the motor frame at one end and to the grounding prong on the grounding-type attachment plug cap at the other end. The grounding conductor shall be identified by a green colored covering or tracer. The attachment plug cap shall conform to the American Standards Association Standard C73. Grounding requirement shall also be applicable when specified or required by Underwriter's Laboratories Inc., Standard No. 73.

3.12.2 *Grounding plug adapter.* When specified (see 6.1), with each grounding plug an adapter shall be furnished to permit its use with a two-prong receptacle.

3.13 Except for type II, classes C and D vacuum cleaners means shall be provided on the vacuum cleaner so that the cord may be coiled, unless the cord may be conveniently wrapped around the vacuum cleaner.

3.14 Dust bag. For type II, classes A and B cleaners the dust bag shall be readily removable, and the joint between the bag and the cleaner shall be dusttight. Exposed bags shall have mechanical means which support the bag clear of the floor when the handle is in the upright operating position. Interior dust bag shall be of the throw-away type.

3.15 Attachments.

3.15.1 *General.* Unless otherwise specified (see 6.1), attachments shall be of nonferrous alloy or steel, except that for type I, class D, and type II, classes A, B, C, and D vacuum cleaners, attachments may be of suitable molded material. The entire exposed surface of each ferrous attachment shall have a corrosion-resistant coating and shall be smooth (see 3.5). Where practicable, each corner and edge shall be rounded to protect the furniture.

3.15.2 *Bare floor tool.* Bare floor tools shall comply with the requirements of table II. Except for type I, class D, model 1 vacuum cleaners and where the tool is an

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Table III.- Requirements for attachments (type I, classes A, B, C and G, and type II, class F vacuum cleaners). [1]

Attachment	Inside diameter (min.)	Length (min.)		Slot			Weight (max.)	
				Width (min.)	Length (min.)	Area (min.)		
	Inches	Feet	Inches	Inch	Inches	Sq. In.	Pounds	Ounces
Bare floor tool	--	1	2	--	13	6-1/2	3	0
Carpet tool	--	--	12	--	11	3	3	0
Radiator and crevice tool	--	--	--	--	--	--	--	--
Floor handle	1-1/4	4	5	--	--	--	3	0
Hose [3]	1-1/2	[2]	--	[2]	--	--	[3]	--
Swivel joint [3]	1-1/2	--	--	--	--	--	2	8

(Type I, class D, and type II, classes A and B vacuum cleaners). [1]

Bare floor tool	--	--	--	--	8	6	3 [4]	0
Carpet tool	--	--	--	--	9	3	2 [4]	0
Radiator and crevice tool	--	--	8	1/4	1-1/8	--	--	12
Hose	1	7	8	--	--	--	--	--
Extension tube	1	4	5	--	--	--	--	--
Hand brush	1-3/4 inches I.D. or 5 inches long							
Upholstery nozzle	--	--	4-1/2	--	--	2	--	8

[1] See 3.16.9 and 3.17.7.
[2] See table I, 3.15.5 and 3.16.8.5.
[3] See 3.16.8.5 and 3.16.8.7.
[4] Not applicable when tool is an integral part of the machine.

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integral part of the cleaners, the tool shall be so constructed that when the tool handle is held stationary the head may be rotated at least 175 degrees about the swivel coupling on the floor handle. Style 1 machines shall swivel 360 degrees.

3.15.3 *Carpet tools.* Carpet tools shall comply with the requirements in table II. Except for type I, class D, model 1 and vacuum cleaners where the tool is an integral part, the carpet tool shall be constructed so that it shall be held in place when rotated 175 degrees about the swivel coupling on the floor handle. Style 1 machines shall swivel 360 degrees.

3.15.4 *Floor handle.* Floor handles shall comply with the requirements in table III. The handle shall be constructed so that the floor tool shall be in proper position on the floor when in use, and shall allow the floor tool to swivel at least 175 degrees. It may consist of one or more sections.

3.15.4.1 *Extension tube.* The extension tube may be either a floor handle or an extension to permit extended reach in floor, wall, or ceiling cleaning. It may be curved or straight as required for the duty to be performed. Length shall be not less than that specified or required for its intended use.

3.15.5 *Hose.* Hose shall be rubber, vinyl, or equally durable plastic, as hereinafter specified. Weight shall be as specified for types I and II (see 3.16.8.5 and 3.17.6.3).

3.15.5.1 *Strength.* When tested as specified in 4.11, there shall be no rupture of any portion of the hose and any resulting diameter shall not differ more than 1/16 inch from the diameter prior to test.

3.15.5.2 *Rubber hose.* The hose shall consist of a rubber lining, reinforced by a helix of spring-steel wire extending the entire length of the hose, except for the connecting ends. The hose shall be completely covered by a durable rubber-impregnated material with nonmarking characteristics. Hose shall conform to the requirements of tables I and III.

3.15.5.3 *Plastic hose.* The plastic hose shall be of vinyl or other equally durable composition and shall be wire-reinforced and nonmarking.

3.15.6 *Couplings.* Couplings shall be arranged so that no metal parts shall come in contact with the floor or furniture. One coupling on the hose shall fit the intake of the cleaner and be of a type which shall not require the use of tools for effective locking or unlocking. A swivel connector shall be supplied between the hose and machine, or between the hose and handle. All attachments shall be readily attached and detached from the opposite end of the hose. This shall include the addition of another length of hose where required. When specified (see 6.1), couplings shall be male.

3.15.6.1 *Strength.* When tested as specified in 4.12, there shall be no rupture or permanent deformation of either the hose or the couplings, and the couplings shall engage and release satisfactorily.

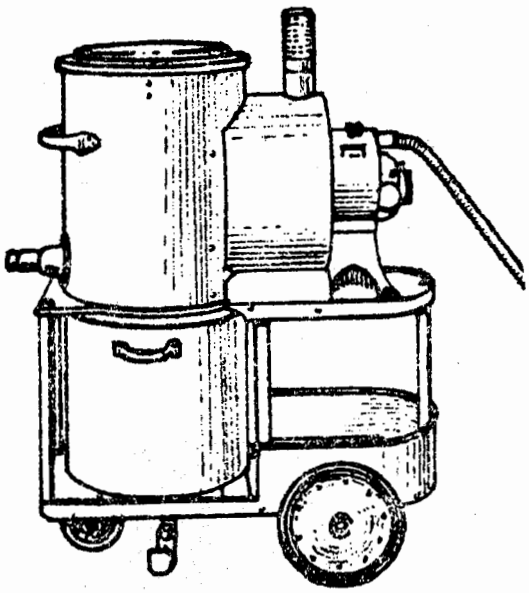
3.16 Type I vacuum cleaners, enclosed filter, suction only.

3.16.1 *General.*

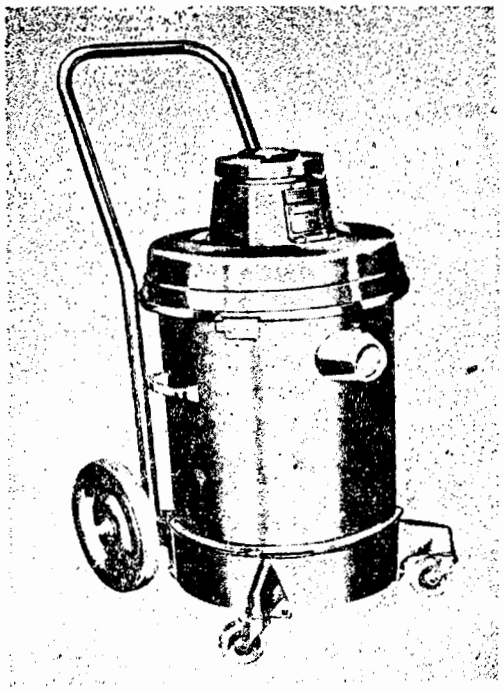
3.16.1.1 *Classes A, B, C, and G.* Type I, classes A, B, C, and G vacuum cleaners shall clean by suction only. Style 1 machines shall be reversible as specified in 3.16.2 and shall comply with the requirements of table I for the class, size, and style specified in the invitation for bids. When specified in the invitation to bid, contract, or order (see 6.1), style 2 machines shall be reversible. Class A, sizes 3 and 4 vacuum cleaners shall be arranged for use with two 50-foot suction hoses connected in parallel.

3.16.1.2 *Class D.* Type I, class D vacuum cleaners shall clean by suction only. Unless otherwise specified (see 6.1), the separator may be of either model 1, horizontal cylinder (for alternate shape see 3.16.1.2.1), or model 2 vertical canister. Models 1 and 2 cleaners shall be arranged for use of paper-bag receptacles and, unless otherwise specified (see 6.1), each machine shall be provided with not less than 12 paper receptacles.

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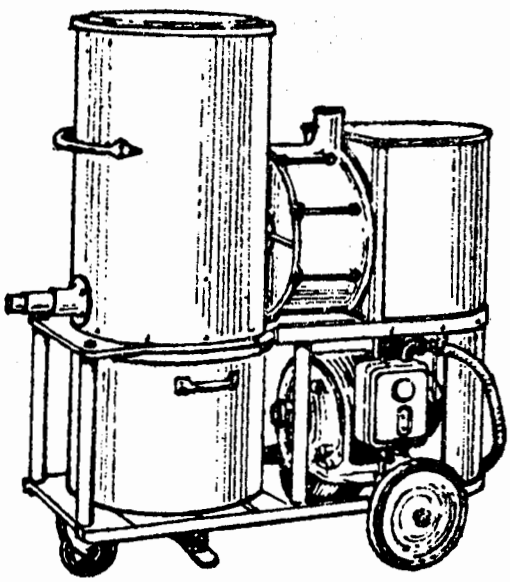


Style 1.

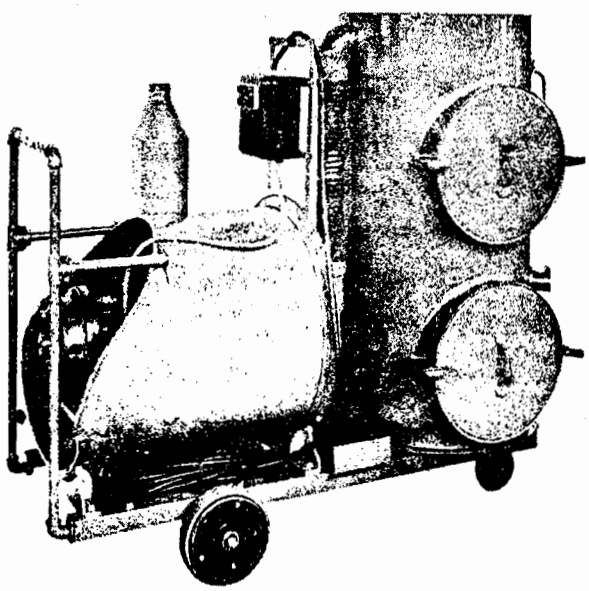


Style 2.

Figure 1. - Type I, class A, size 1, 2000 cu. in. minimum enclosed separator, suction only.



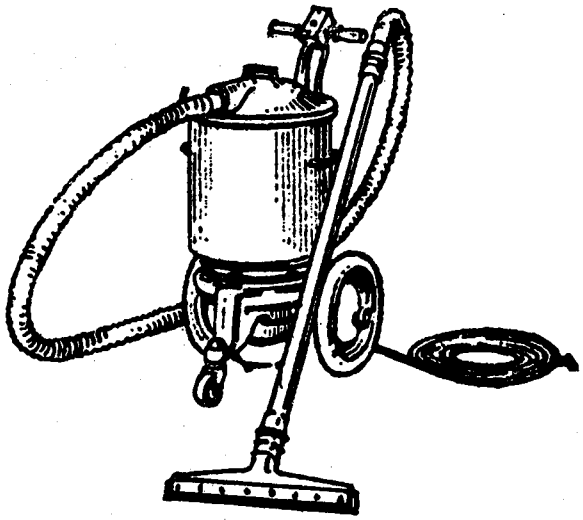
With removable receptacle.



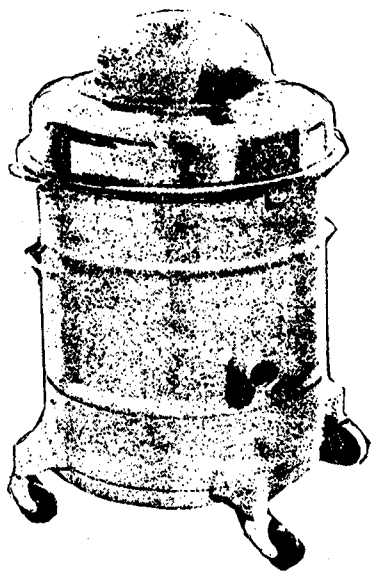
Without removable receptacle.

Figure 2 - Type I, class A, size 2, 3 and 4, 2000 cu. in. minimum and 3200 cu. in. minimum enclosed separator, suction only.

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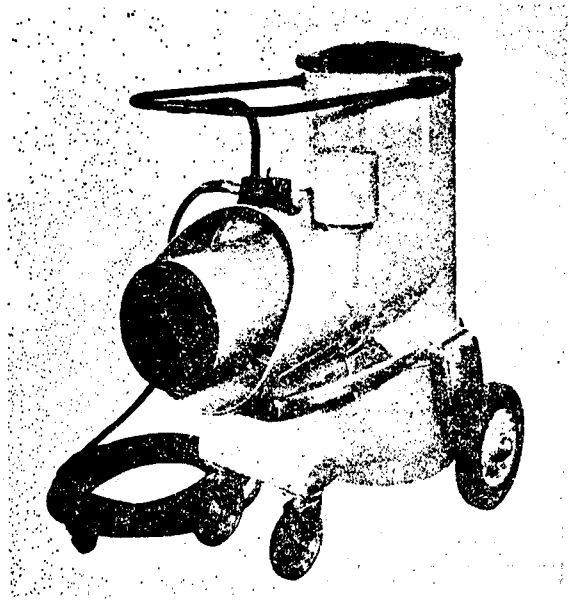
Style 1.



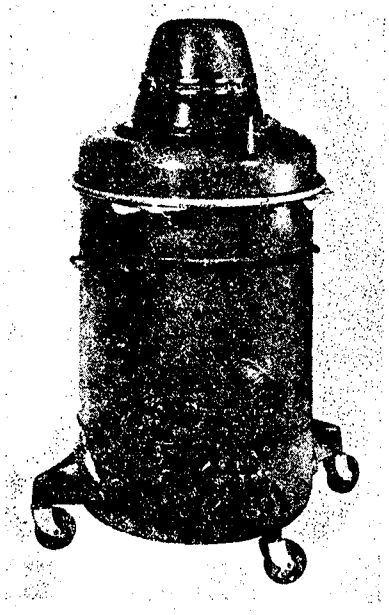
Style 2.

Figure 3. - Type I, class B, size 1, 690 cu. in. minimum enclosed separator, suction only.

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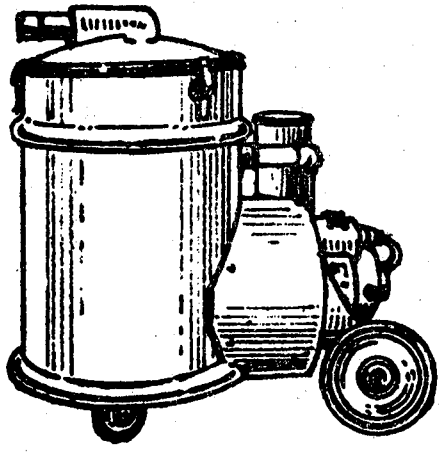


Style 1.

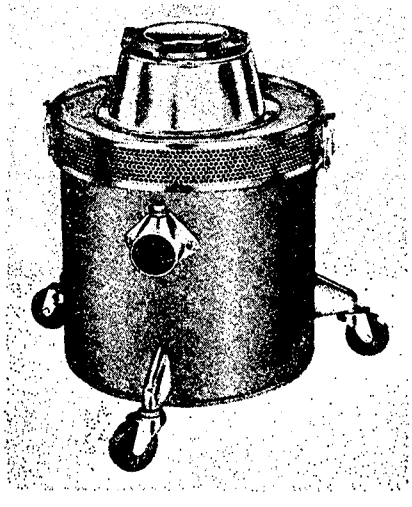


Style 2.

Figure 4. - Type I, class B, size 2, 690 cu. in. minimum enclosed separator, suction only.



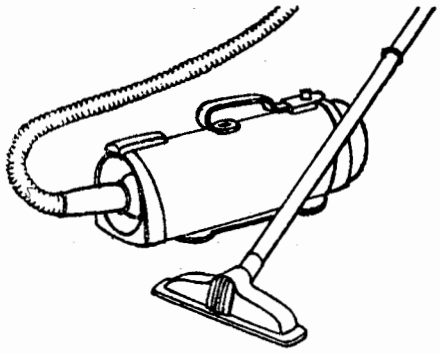
Style 1.



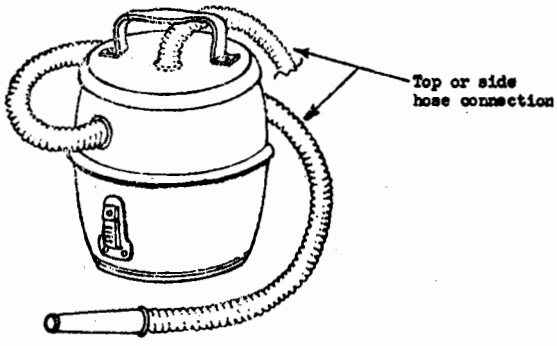
Style 2.

Figure 5. - Type I, class C, size 1, 360 cu. in. minimum enclosed separator, suction only.

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Model 1. - Horizontal cylinder.



Model 2. - Vertical canister.

FIGURE 6.—Type I, class D, household, enclosed separator, suction only.

3.16.1.2.1 Model 1 horizontal cylinder cleaner shall be interpreted as one with its major axis horizontal. It is neither required that the receptacle be a true cylinder nor that its cross section be round in shape.

3.16.2 *Reversibility.* Each type I, classes A, B, and C, style 1 vacuum cleaner shall function either as a blower or as an exhauster. In either case, the air shall pass through the filters so that no dirt will be blown into the room.

3.16.3 *Mounting.* Type I and type II, class F cleaners shall be supported on wheels or casters mounted on a chassis or on the tank. The wheels and casters shall have nonmarking, natural or synthetic rubber tires and ball, roller, or self-lubricating sleeve bearings. The mounting shall consist of one of the following arrangements:

Arrangement 1—One or more swivelling wheels or casters and 2 nonswivelling 6-inch-minimum-diameter wheels. At least one wheel or caster shall be fitted with a brake.

Arrangement 2—Three or more 2-inch-minimum-diameter wheels or casters, at least one of which swivels. At least one wheel or caster shall be fitted with a brake.

Arrangement 3—One or more swivelling wheels or casters with 3-inch-minimum diameter, and two nonswivelling 8-inch minimum diameter wheels. At least one wheel or caster shall be fitted with a brake.

3.16.3.1 *Arrangements 1 and 3.* Arrangements 1 and 3 mounting shall provide for easy maneuverability and operation over the most irregular floors found in government buildings containing such impediments as floor irregularity and 4-inch-high sills. Wheels and mounting shall be of a design to assist in portability of cleaner on stairs, risers, and over deep carpets.

3.16.3.2 *Arrangement 2.* Arrangement 2 mounting shall perform as specified for arrangement 1 except for the requirement for overcoming 4-inch-high sills and mounting stairs.

3.16.3.3 Type I cleaners, except style 2, class D and class G shall be provided with arrangement 1 mounting. Type I, style 2 and type II, class F cleaners, unless otherwise specified, shall be provided with either arrangement 1 or arrangement 2 mounting. Type I, class G cleaners shall be provided with arrangement 3 mounting.

3.16.3.4 *Mounting for class D cleaners* Type I, class D cleaners may be mounted on wheels, casters, polished metal runners, or other equally smooth surface for mobility of the cleaner.

3.16.4 *Handles or handholds.* Handles or handholds shall be provided on all type I and type II, class F cleaners for easy movability. These may consist of a handle of tubular, U-shaped, corrosion-resistant metal, or metal component handholds of adequate strength, fitted and securely attached to the cleaner.

3.16.5 *Vacuum producer.* The vacuum producer for type I, classes A, B, C, and G

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vacuum cleaners shall be of the multiple-turbine type or single-stage fan-blade design, or of the rotary-pump positive-displacement type. The rotary-pump positive-displacement type shall be so designed that no dirt is pulled through the vacuum producer. The impellers of the rotary-pump type producers shall be adequately keyed to ground steel shafts mounted on ball bearings, and suitably geared together.

3.16.5.1 *Type I, class G.* The vacuum producer for type I, class G cleaners shall be mounted on top of the dirt receptacle and shall be removable. The top shall be lined with sound-deadening material and equipped with a felt exhauster. The exhaust air shall be diffused around the entire circumference of the machine, to minimize turbulence; and shall be directed away from the floor. The vacuum producer shall provide an air-intake volume of not less than 145 c.f.m. measured through a 5/8-inch orifice at the hose connection of the machine and with both disposable filter and cloth filter in use.

3.16.6 *Receptacle and filter.* The dirt shall fall into a suitable receptacle which can be readily cleaned. There shall be a filter or filters in the vacuum cleaner which shall be readily cleanable.

3.16.6.1 *Type I, class G.* The dirt receptacle of type I, class G cleaners shall have a minimum recoverable capacity of 5 gallons of liquid or 2/3 bushel of dirt. The tank shall be of seamless-steel construction having no welds or angles which could retain dirt or soil. The interior shall be finished with a plastic resin material which will resist corrosion and oxidation. Air through the hose shall be filtered through a minimum of two filters. The first (primary) filter shall consist of a disposable-paper type having openings of not more than one micron in size. The second (secondary) filter shall be constructed of mitten-flannel cloth and shall have an area of not less than 400 square inches and shall be provided with a suitable frame to insure full utilization of the surface area.

3.16.7 *Motor.* Motors for type I cleaners shall be as specified in 3.16.7.1 through

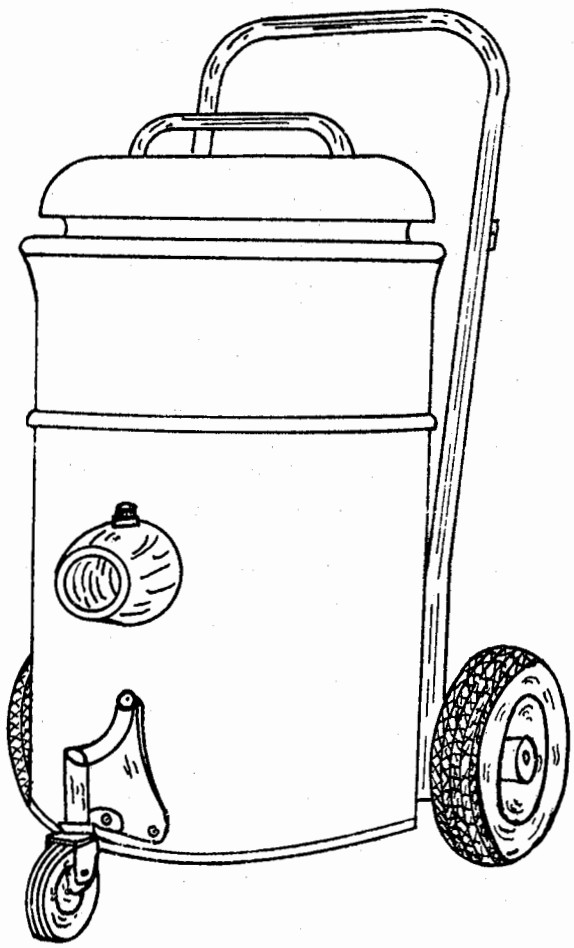


FIGURE 7.—Type I, class G, size 1, hospital, enclosed separator, suction only.

3.16.7.4.

3.16.7.1 *Duty.* Motors shall be of the continuous-duty type, 8 hours per day for classes A, B, C, and G machines. The motors shall be either one-hour varying duty or continuous duty, 8 hours per day, for class D machines.

3.16.7.2 *Cooling.* For type I, classes A, B, and C, style 1 vacuum cleaners, the motor shall be capable of developing the brake horsepower specified in table I without the use of air from the vacuum producer for cooling purposes.

3.16.7.2.1 *Type I, class G.* The motor(s) for type I, class G cleaners shall be cooled by air that is drawn in the suction end of the hose and passes through all filters of the machine. A monitor switch shall be

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mounted within the vacuum producer and connected to a thermostat which in turn shall be connected to the motor(s). The thermostat shall be set to cutoff when the motor(s) reach a temperature of 180°F. The switch shall have three operating positions, namely "ON", "OFF", and "RESET".

3.16.7.3 *Motor brushes.* The service life of the motor brushes under normal working conditions shall be not less than the following values: For classes A, B, C, and G, 700 hours; for class D, 500 hours. For brush life test purposes the normal working condition shall be simulated by adjusting the amount of cleaning air entering the cleaner so that the watts input to the cleaner will be equal to the average of the watts input when the cleaner is operating under sealed and wide open conditions of the suction air intake.

3.16.7.4 *Motor bearings.* The service life of the motor bearings under normal working conditions shall be not less than the following values: For classes A, B, C, and G, 2,000 hours; for class D, 1,500 hours.

3.16.8 *Attachments.* The attachments for type I cleaners shall comply with the requirements of 3.15 and tables I and III. Cleaners shall be furnished with or without attachments and hose, as specified in the invitation for bids (see 6.1).

3.16.8.1 *Bare floor tool.* The wearing surfaces of bare floor tools or attachments thereto for type I, classes A, B, C, and G vacuum cleaners shall be either rubberized fabric, felt, or bristles.

3.16.8.2 *Carpet tool.* The wearing surfaces shall be of a nonferrous alloy or molded material.

3.16.8.3 *Radiator and crevice tool.* Radiator and crevice tool for type I, class D vacuum cleaners shall be of the material specified in 3.15.1 or of vulcanized fiber.

3.16.8.4 *Floor handle.* Floor handles for type I, classes A, B, C, and G vacuum cleaners shall have a free-acting swivel joint at the tool end.

3.16.8.5 *Hose.* For type I, classes A, B, C, and G vacuum cleaners the weight of a 25-foot length of hose with couplings shall be not more than 16 pounds. The length of any piece of uncoupled hose shall be not less than 10 feet and shall not exceed 25 feet. The hose for type I, class G cleaners shall be in accordance with 3.15.5.3.

3.16.8.6 *Couplings.* For type I, classes A, B, C, and G vacuum cleaners there shall be means for coupling at each end of each length of hose. For type I, class D vacuum cleaners, the hose shall have a swivel joint at the machine end and suitable means for applying attachments at the other end.

3.16.8.7 *Swivel joint.* Type I, style 1 and type I, class A, sizes 3 and 4 vacuum cleaners shall be provided with a ball-bearing swivel elbow joint. The elbow joint shall connect the floor handle with the hose so as to permit the handle to move freely and prevent strain at the handle.

3.16.8.8 *Tool fittings.* All tools for type I, class G cleaners shall be equipped with male fittings.

3.16.9 When attachments are specified for type I cleaners (see 6.1), they shall be provided as follows (unless a variation from this list is specified):

Cleaner classification	*Attachments*
Type I, class A Size 1, style 1 Size 2 Type I, class B Size 1, style 1 Size 2, style 1 Type I, class C Size 1, style 1	1 carpet tool 1 bare floor tool 1 radiator and crevice tool 1 floor handle 1 hose swivel joint Hose
Type I, class A Size 3 Size 4	2 carpet tools 2 bare floor tools 2 radiator and crevice tools 2 floor handles 2 hose swivel joints Hose
Type I, class A Size 1, style 2 Type I, class B Size 1, style 2 Size 2, style 2 Type I, class C Size 1, style 2	1 carpet tool 1 bare floor tool 1 radiator and crevice tool 1 floor handle Hose

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Cleaner classification (Cont'd)	*Attachments*
Type I, class D	1 carpet tool 1 bare floor tool 1 radiator and crevice tool 1 floor handle 1 upholstery tool Hose
Type I, class G	1 carpet tool 1 bare floor tool 1 radiator and crevice tool 1 floor handle 1 round dusting brush, 4 inches 1 floor brush, 10 inches 1 water pick-up tool, 16 inches, with replaceable rubber blades and spot brush attachment 1 water separator with positive shutoff float Hose

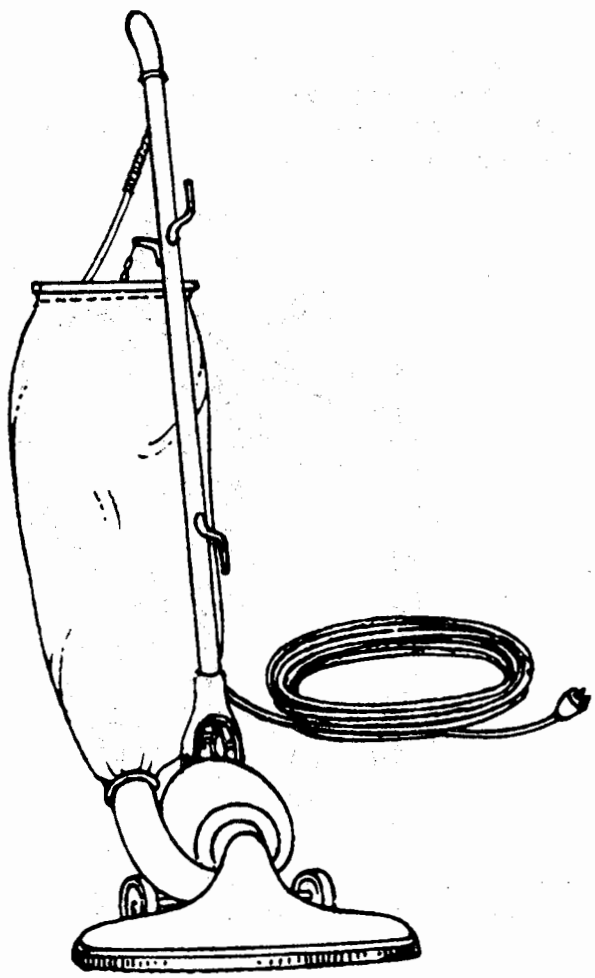


FIGURE 8.—Type II, class A, floor model, exposed separator, suction only.

3.17 Type II vacuum cleaners, exposed filter. Unless otherwise specified (see 6.1), for classes A and B the handle shall be adjustable to an approximately horizontal position and the overall height of cleaner housing shall not exceed 7 inches. Also the dust bags shall consist of an outer container and an inner throw-away filter.

3.17.1 *General.*

3.17.1.1 *Class A.* The type II, class A vacuum cleaner shall clean by suction only. It shall be a floor model without agitator, and shall comply with the requirements of table I. The machine shall be suitably arranged so that the nozzle shall be in proper position for cleaning any type of carpet. This may be accomplished either automatically or by manual adjustment.

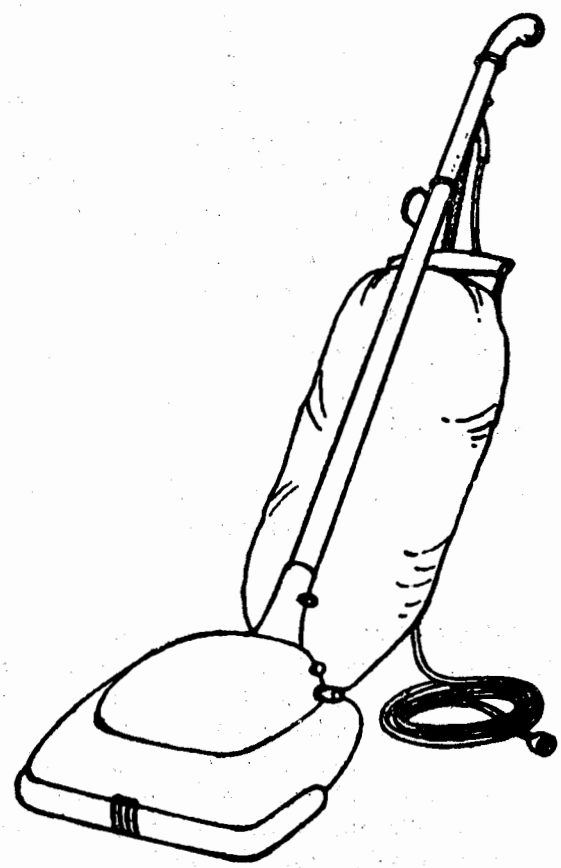


FIGURE 9.—Type II, class B, floor model, suction and power-driven agitator, exposed separator.

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3.17.1.2 *Class B.* The type II, class B vacuum cleaner shall clean by combined suction and a removable power-driven agitator. The cleaner shall be a floor model and shall comply with the requirements of table I.

3.17.1.3 *Class C.* The type II, class C vacuum cleaner shall clean by suction only. It shall be a hand model and shall comply with the requirements of table I. A scrap trap shall be provided which will prevent metallic and heavy debris from entering the impeller case of the cleaner.

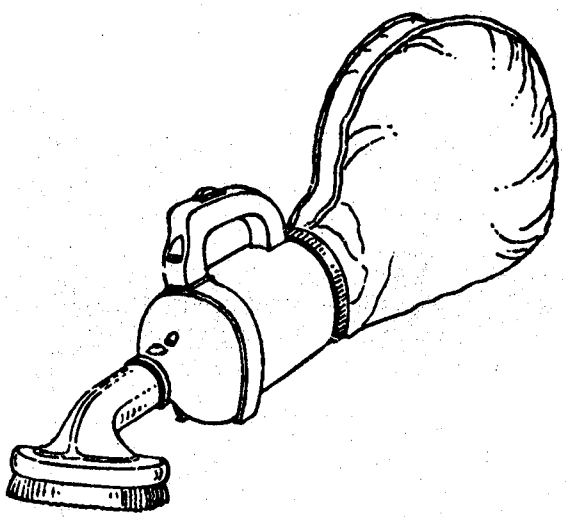


FIGURE 10.—Type II, class C, hand model, suction only.

3.17.1.4 *Class D.* The type II, class D vacuum cleaner shall clean by combined suction and removable power-driven agitator. The cleaner shall be a hand model and shall comply with the requirements of table I. A scrap trap shall be provided which will prevent metallic and heavy debris from entering the impeller case of the cleaner.

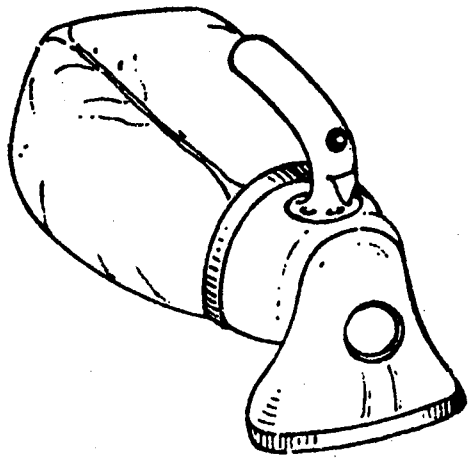


FIGURE 11.—Type II, class D, hand model, suction and power-driven agitator.

3.17.1.5 *Class F.* The type II, class F vacuum cleaner shall clean by suction only. It shall be a tank model and shall comply with the requirements of table I.

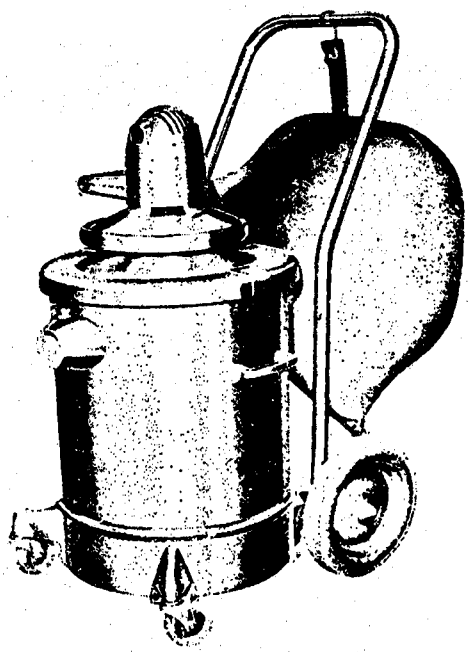


FIGURE 12.—Type II, class F, sizes 1 and 2, floor model, 1200 cu. in. minimum.

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3.17.2 *Mounting.* Type II vacuum cleaners, except for hand models, classes C and D, shall be mounted on smooth wheels or casters which will not scratch or injure the floor or floor covering. For class F see 3.16.3 and 3.16.3.3.

3.17.2.1 *Class A.* Each type II, class A vacuum cleaner shall consist of a nozzle, a removable brush fixed to the nozzle lip, a vacuum producer, a handle of metal or fiber tubing for pushing the cleaner, a motor, a switch, a cord, and a dust bag, assembled together and mounted as a unit on not less than three wheels or casters.

3.17.2.2 *Class B.* Each type II, class B vacuum cleaner shall comply with the requirements of 3.17.2.1, except as follows: A fixed brush is not required, a metal handle is required for size 2 vacuum cleaners, and size 2 vacuum cleaners shall be mounted on not less than four ball or metal sleeve bearing wheels that are smooth and securely attached.

3.17.2.3 *Class C.* Each type II, class C vacuum cleaner shall include a nozzle without agitator, a vacuum producer, hand grip or handle for supporting and manipulating the cleaner, motor, switch, cord, dust bag, and a removable brush fixed to the nozzle lip mounted as a unit.

3.17.2.4 *Class D.* Each type II, class D vacuum cleaner shall include a nozzle with power-driven agitator, a vacuum producer, handle, motor, switch cord, and a dust bag mounted as a unit.

3.17.2.5 *Class E.* Not applicable.

3.17.2.6 *Class F.* Each type II, class F vacuum cleaner shall consist of a motor and vacuum producer mounted on a metal tank, a switch, cord, dust bag and attachments (see 3.17.6). The tank shall be of rigid metal construction and shall be adapted to receive heavy material; the light dust passing through the tank shall be collected in the tank and prevented from reaching the motor by means of an interposing filter bag. The tank shall be mounted in conformance with 3.16.3 and 3.16.3.3. When specified (see 6.1), tanks of other than metal may be used.

3.17.3 *Nozzle.* Unless the vacuum cleaner automatically assumes proper position on the floor covering to give effective performance, there shall be a positive means for adjusting the distance between the nozzle and the floor for type II, classes A and B vacuum cleaners. There shall be a bumper of suitable material on the front and sides of the nozzle.

3.17.4 *Agitator.* For type II, classes B and D vacuum cleaners an agitator shall be in the nozzle and rotated by the motor. The agitator shall have a brush(s), or suitable metal surfaces, metal edges, or a combination of the foregoing which contact the material to be cleaned. If a vacuum cleaner is operated on a carpet, devices on the surface of the agitator shall agitate the carpet. Devices which may wear appreciably, such as bristles, shall be replaceable either by securing new bristles in the sockets of the agitator or by replacing the bristle units, at the option of the contractor. Type II, class B, size 2 vacuum cleaners, in addition to above requirements, shall have adjustment to permit the agitator to be lowered as the bristles wear.

3.17.4.1 *Agitator bearings.* If the agitator bearings are readily removable, the service life of the bearings shall be not less than the service life of the agitating device. If the bearings are not readily removable, the service life of the bearings shall be not less than the service life of the cleaner. Type II, class B, size 2 vacuum cleaners shall have

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readily removable ball bearings, grease packed and dirt sealed. The service life of the bearing shall not be less than the service life of the agitator.

3.17.5 *Motor.* Motor for type II cleaners shall be as specified in 3.17.5.1 through 3.17.5.3.

3.17.5.1 *Duty.* Motors shall be either one-hour varying duty or continuous duty type for classes A, B (size 1), C, and D cleaners. The motors shall be continuous duty type for class B (size 2) and class F.

3.17.5.2 *Motor brushes.* The service life of the motor brushes under normal working conditions shall be not less than the following values: For class A, 500 hours; for class B, size 1, 500 hours; for class B, size 2, 700 hours; for classes C and D, 300 hours; for class F, 700 hours. For brush-life test purposes the normal working condition shall be simulated by adjusting the amount of cleaning air entering the cleaner so that the watts input to the cleaner will be equal to the average of the watts input when the cleaner is operating under sealed and under wide open conditions of the suction air intake.

3.17.5.3 *Motor bearings.* The service life of the motor bearings under normal working conditions shall be not less than the following values: For classes A and B, 1,500 hours; for classes C and D, 600 hours; and for class F, 2,000 hours.

3.17.6 *Attachments.* Type II, classes A, B, and F vacuum cleaners shall be furnished with or without attachments as specified herein and in the invitation for bids (see table II and 6.1).

3.17.6.1 *Radiator and crevice tool.* Radiator and crevice tools for type II, classes A and B vacuum cleaners shall be of the material specified in 3.15.1, or of vulcanized fiber.

3.17.6.2 *Hand brush.* Hand brushes for type II, classes A and B vacuum cleaners shall rotate freely about the axis of the hose coupling or tool coupling. The brush portion of the tool shall be readily removable. The brush shall be equipped with a rubber bumper of new, natural or synthetic rubber.

3.17.6.3 *Hose.* For type II classes A and B, hose may be lightweight rubber or plastic meeting the strength test specified in 3.15.5.1.

3.17.6.4 *Couplings.* For type II, classes A and B vacuum cleaners, hose shall have a swivel joint at the machine end and suitable means for applying attachments at the other end. For type II, class F vacuum cleaners there shall be means for coupling at each end of each length of hose.

3.17.6.5 *Extension tube.* Extension tube for type II, classes A and B vacuum cleaners shall be of not more than two-piece construction.

3.17.7 When attachments are specified for type II, classes A, B, and F cleaners (see 6.1), they shall be provided as follows (unless a variation from this list is specified):

Type II, class A:
- 1 hose (7 feet and 8 inches minimum length).
- 1 radiator and crevice tool.
- 1 extension tube.
- 1 hand brush.
- 1 upholstery tool.

Type II, class B:
- 1 hose (7 feet and 8 inches minimum length).
- 1 radiator and crevice tool.
- 1 extension tube.
- 1 hand brush.
- 1 upholstery tool.

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Type II, classes C and D:

No attachments required.

Type II, class F:

1 carpet tool.
1 bare floor tool.
1 radiator and crevice tool.
1 floor handle.
1 hose.

3.18 **Workmanship.** Workmanship shall be first class.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for inspection and acceptance.** Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1 *Component and material inspection.* In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification.

4.2.2 *End item inspection.*

4.2.2.1 *Visual examination.* Examination of the end item shall be in accordance with the classification of defects and acceptable quality levels (AQL's) set forth in table IV. The lot sizes shall be expressed in units of vacuum cleaners for the purpose of determining the sample size in accordance with MIL-STD-105. The sample unit for this examination shall be one completely fabricated vacuum cleaner of each type, class, and size offered for delivery at one time.

TABLE IV.—*Classification of defects*

Examine	Defect	Major	Minor
Finish	Not finished as specified		X
Design	Any characteristic not in accordance with the specific requirements ...	X	
	Not type, class size specified ...	X	
Construction & workmanship	Fractured, split, bowed, malforme or otherwise impaired	X	
(General, applicable to all components)	Parts missing or not specified type or size	X	
	Sharp burr, sliver or splinter that may be injurious to personnel	X	
	Part misplaced, loose or not in proper alignment	X	
Electrical assembly	Not furnished wit three conductor cords and plugs (when specified) .		X
	Not furnished wit green colored ground conductor (when specified) .		X
	Wiring cut, abraded not properly joined or loose at terminals	X	
	Adequate slack not provided to relieve strain on wiring .		X
Lubrication (where applicable)	Not lubricated where required ...	X	
	Insufficiently lubricated		X

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TABLE IV.—*Classification of defects (cont'd)*

Examine	Defect	Major	Minor
Marking—identification, instructions, manual, et cetera	Missing, incomplete, not legible, not specified type or size, not in proper location, not affixed in the prescribed manner, not correct color, plate, label or paper; not fabricated of the specified material, not stenciled, stamped, embossed, printed, lithographed, et cetera, whichever required		X

4.2.2.2 *Dimensional examination.* Inspection shall be made of the end item to determine compliance with dimensions specified. Any deviation from specified dimensions shall be classified as a defect.

4.2.2.3 *Inspection levels and AQL's.* The acceptable quality levels (AQL'S) expressed in defects per hundred units and inspection levels (for sample size), shall be as follows:

For examination in	Inspection Level	AQL's Major	AQL's Total
Table IV	II	4.0	6.5
4.2.2.2 (one class) .	L4	—	6.5

4.3 **End item testing.**

4.3.1 *Group "A" tests.* A random sample of vacuum cleaners shall be selected from each inspection lot and subjected to each of the tests listed below, using inspection level II of MIL-STD-105 and an acceptable quality level (AQL) of 4.0 defects per hundred units.

Group "A" Tests	*Reference Par.*
Power input	4.4.1
Operating test	4.4.1
Dielectric strength	4.4.2
Insulation test	4.4.3

4.3.2 *Group "B" tests.* A random sample of vacuum cleaners shall be selected from each inspection lot and subjected to each of the tests listed below using inspection level L8 of MIL-STD-105 and an acceptable quality level (AQL) of 4.0 defects per hundred units.

Group "B" Tests	*Reference Par.*
Heating (temperature rise)	4.5
Exposed dust bag test	4.9
Suction tests	4.10
Hose test	4.11
Coupling test	4.12

4.3.3 *Group "C" tests.* Testing shall be performed on one vacuum cleaner from each inspection lot to determine compliance with the requirements for Group "C" tests, with no evidence of failure.

Group "C" Tests	*Reference Par.*
Life test (10 hours)	4.4.4
Switch tests	4.7
Cleaning attachments	4.8

4.3.4 *Radio interference reduction.* Radio interference suppression tests shall be performed to determine compliance with 3.10.1 and 4.6.3.

4.3.5 *Examination of preparation for delivery.* An examination shall be made to determine that the requirements for preservation, packaging, packing and marking as required by section 5 of this specification are complied with. Defects will be scored in accordance with table V. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level (sample size) shall be level L4 and the acceptable quality level (AQL) shall be 4.0 defects per hundred units.

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TABLE V.—*Classification of defects*

Examine	Defect
Markings (exterior and interior container)	Incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any nonconforming component; component missing, damaged or otherwise defective affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling, bulging or distortion of containers.
Contents (exterior and interior containers)	Number per container is more or less than required.
Weight (exterior and interior containers)	Gross weight exceeds requirements.
Preservation	Preservation improperly applied or missing, strapping, sealing, braces, blocking and cushioning inadequate, improper or missing; bolts and closures improperly affixed; components improperly secured.

4.4 Electrical tests of complete vacuum cleaners.

4.4.1 *Operating tests.* Operating tests on universal motors shall be made by using 115 volts direct current and/or 115 volts, 60-cycle alternating current, as required by the purchaser. Other motors shall be tested with rated voltage, frequency and current.

4.4.2 *Dielectric strength.* Sample vacuum cleaners shall be tested for dielectric strength. With the motor mounted on the vacuum cleaner and at the maximum operating temperature, the insulation shall not fail (break down) when the specified test potential is applied for not less than one minute between the parts which carry current and any metallic part of either the motor or the cleaner which does not carry current. The test shall be made with 60-cycle alternating current. For motors not exceeding 1/2 horsepower, the test potential shall be not less than 900 volts. For motors exceeding 1/2 horsepower, and for capacitors not mounted within the motor housing, the test potential shall be not less than twice the rated voltage plus 1,000 volts. Capacitors mounted within the housing shall be tested with the same voltage as that specified for the respective motors.

4.4.3 *Insulation resistance.* All vacuum cleaners shall be tested for compliance with 3.9.2.

4.4.4 *General compliance.* Except for type II, classes C and D, hand model vacuum cleaners, each sample cleaner shall be operated in the condition received, continuously for 10 hours with orifice wide open. The cleaners shall not be lubricated during this run and shall have adequate lubricant in the bearings at the end of the run. At the end of this run, commutator motors shall operate with practically sparkless commutation. Practically sparkless commutation shall be understood to mean an entire absence of sparking other than fine pin-point blue sparks along the edges of the brushes (heavy sparks or yellow sparks will not be accepted). After the foregoing test, and also for type II, classes C and D vacuum cleaners, the sample vacuum cleaners shall be tested to determine compliance with requirements for motors and suction as specified in table I.

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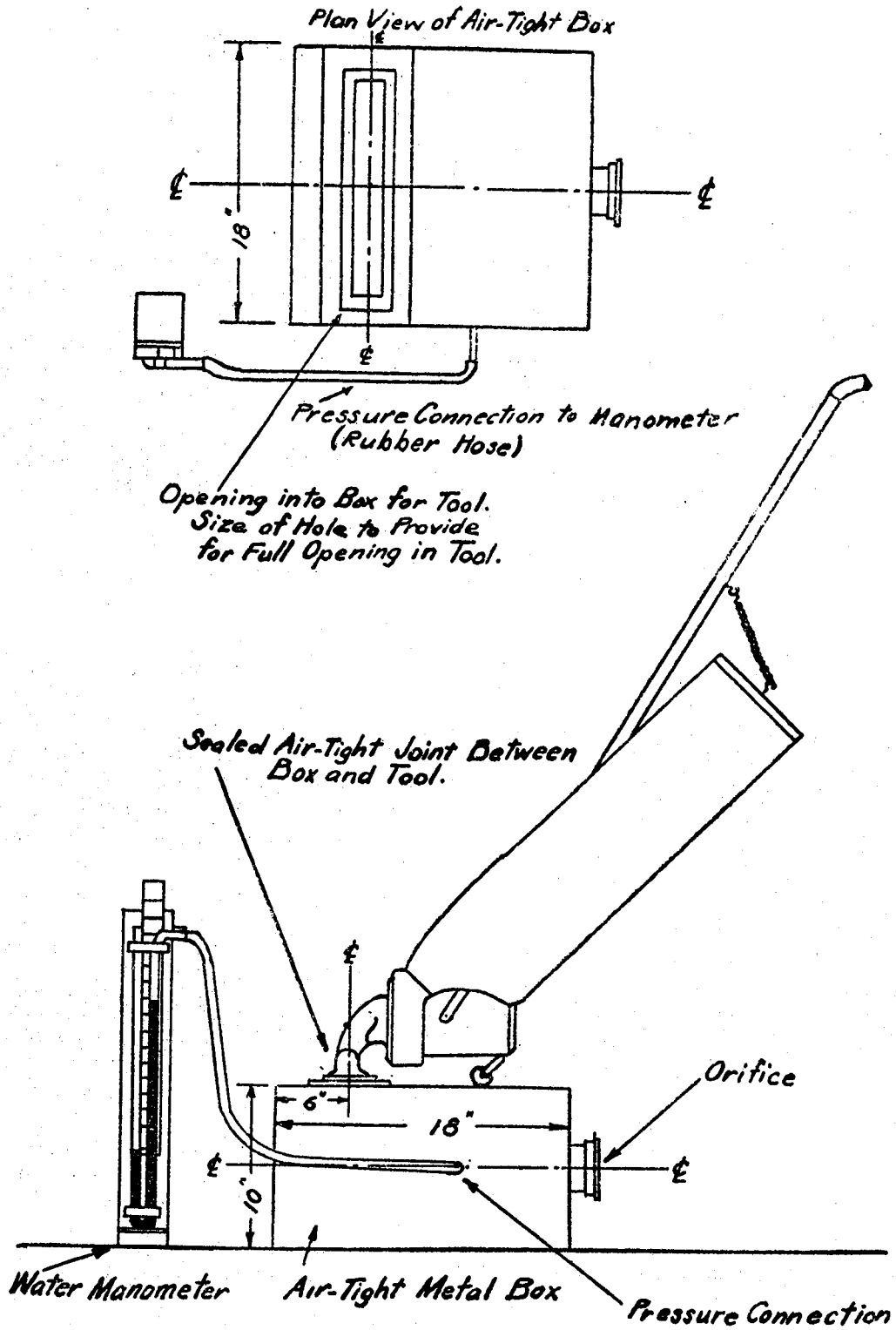


FIGURE 13.—Apparatus for suction test.

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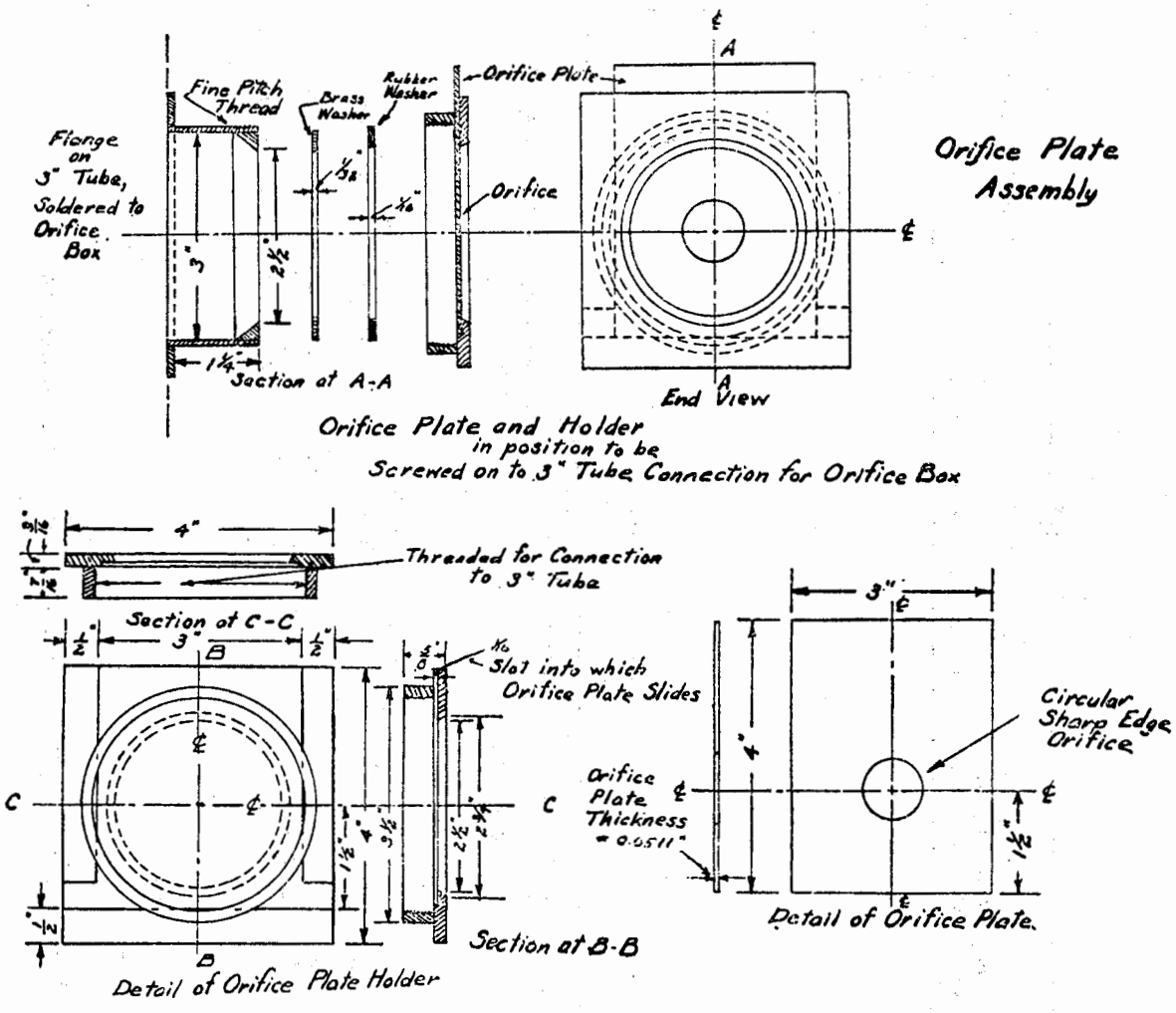


FIGURE 14.—Method of attaching orifice to air meter in suction test.

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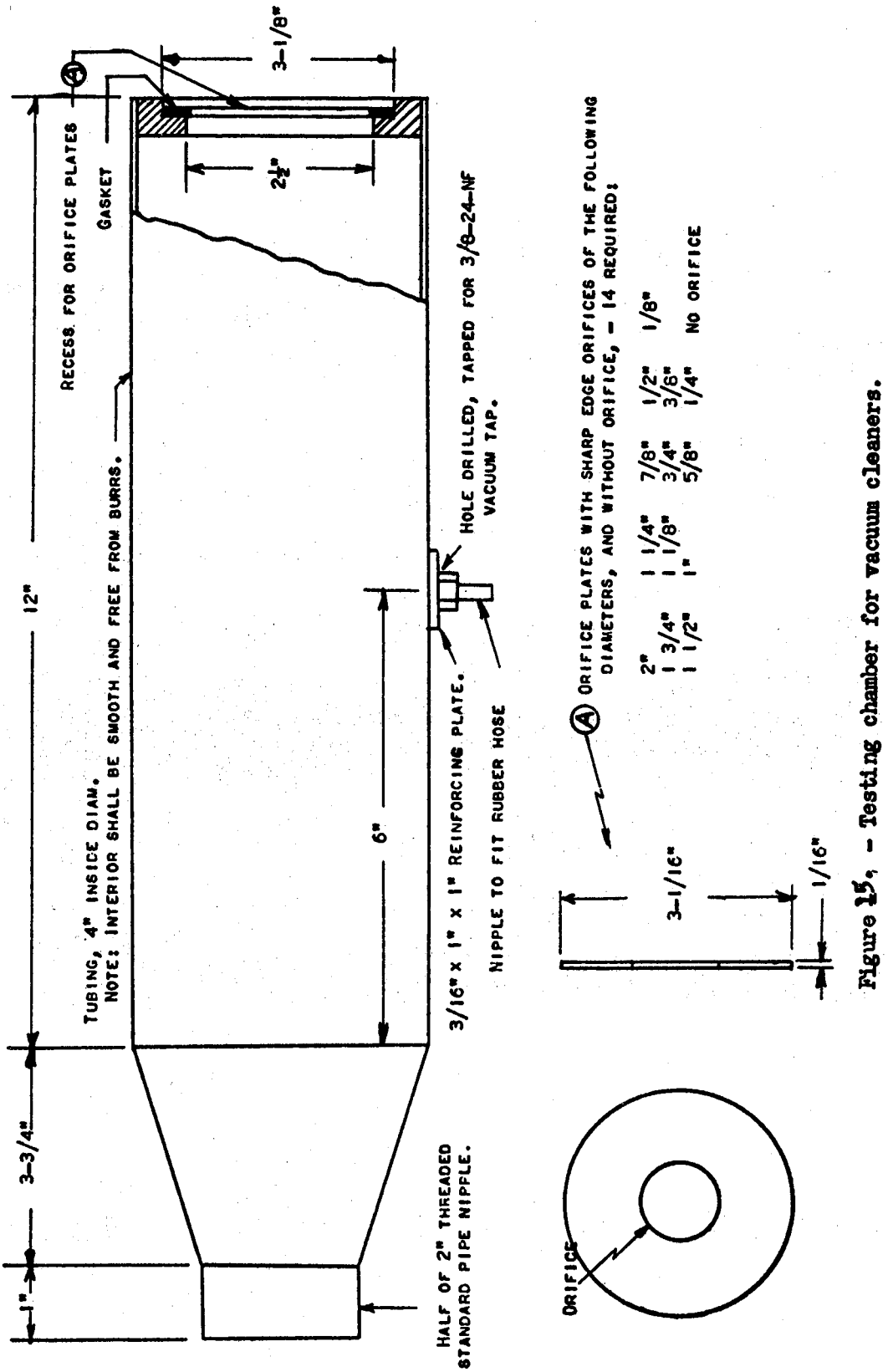


Figure 15. - Testing chamber for vacuum cleaners.

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4.5 Motor temperature rise. The motor of the sample cleaner of each type, class, size and style shall be operated at full rated voltage at a normal ambient temperature of 25°C., unless otherwise specified, up to a maximum of 40°C. The machine shall be operated with both 5/8-inch and 7/8-inch orifice wide open, for continuous periods of two hours each except that for type II, classes A, B, C and D vacuum cleaners, the continuous periods shall be one hour each. At the end of the continuous run periods, the observable temperature rise of motor windings shall not exceed 50°C. for protected motors, and 65°C. for universal and enclosed motors. Temperature measurements shall be made by means of thermometers or thermocouples. The 7/8-inch orifice test shall follow immediately after the 5/8-inch orifice test. This test does not relieve the manufacturer from furnishing motors meeting the requirements of 3.9 through 3.9.6 and 3.16.7 through 3.16.7.4.

4.5.1 When class H insulation is used, the allowable temperature rise shall be 155°C. for universal and enclosed motors. Temperature measurements shall be made in accordance with the methods described in 4.5.

4.6 Motor parts performance test.

4.6.1 *Motor brushes.* If required, motor brushes shall be tested for conformance with the service-life requirements in 3.16.7.3 and 3.17.5.2.

4.6.2 *Motor bearings.* If required, motor bearings shall be tested for conformance with the service-life requirements in 3.16.7.4 and 3.17.5.3.

4.6.3 *Radio interference reduction.* Unless otherwise specified (see 6.1), one of each type, class, size and style of vacuum cleaner radio-interferenced suppressed in accordance with 3.10.1, will be tested as described in the applicable specification by authorized representatives of the contracting officer. Upon approval by the contracting officer, and provided all other requirements of the specification are met, the vacuum cleaner shall be used as a model for all production units.

4.7 Switch tests. A switch or other device which controls the motor, unless recognized as being suitable for the application or unless so interlocked that it will never have to break the locked-rotor motor current, shall be capable of performing acceptably when subjected to an overload test consisting of 50 cycles of operation, making and breaking the locked rotor current of the appliance. There shall be no electrical or mechanical failure of the device or undue pitting or burning of the contacts.

4.8 Cleaning attachments tests. A test shall be made with each kind of cleaning attachment furnished. The tool shall be attached to the length of hose specified in table I and furnished with the machine, and under normal working conditions each tool shall satisfactorily perform the work for which it is designed.

4.9 Exposed dust-bag test. The vacuum cleaner under test, if fitted with dust bag, shall be operated for 2 minutes doing ordinary cleaning, after which time a thin piece of damp muslin about 8 inches square shall be placed on the central portion of the upper surface of the bag and kept there for one minute while the cleaner is in operation. The muslin shall not be appreciably soiled.

4.10 Suction tests. The suction requirements specified for the respective types, classes, sizes and styles shall be determined on the sample vacuum cleaners as described hereinafter (see 4.10.1). Cleaners having power-driven agitators shall be tested with the agitator disconnected. Tests shall be conducted at a temperature of approximately 68°F. and barometric pressure of approximately 29.00 to 30.00 inches. Suction requirements shall be met without overloading the motor. The voltage, phase and frequency at the motor connections during the test shall be as specified for the performance in table I.

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4.10.1 *Type I, classes A, B, C, D, and G, and type II, class F vacuum cleaners.* The apparatus used shall be as shown in figure 15, or a hollow metal globe, 4 inches in diameter. The apparatus shall be attached to the end of a 25-foot length of hose (furnished with type I, classes A, B and C, and type II, class F vacuum cleaners), and at the end of 7-foot 8-inch length of hose (furnished with type I, class D vacuum cleaners), and fitted with the required size of sharp-edged orifice. The vertical distance between the water levels in the two columns of a manometer connected at approximately the center of the apparatus or globe diameter shall be measured to the nearest 0.10 inch for determining the suction of the vacuum cleaner.

4.10.2 *Type II, classes A, B, C, and D vacuum cleaners.* The apparatus used shall be as shown in figure 13. The air meter shall be of the orifice type, using sharp-edged orifices in plates 0.0511 inch in thickness. The method of attaching the orifice to the air meter shall be as shown in figure 14.

4.11 **Hose test.** A compressive load of not less than 200 pounds shall be applied to any portion of the hose by means of parallel surfaces having a length of not more than 3 inches parallel to the axis of the hose, for not less than 2 minutes, and then the load released. There shall be no rupture of any portion of the hose and any resulting diameter shall not differ more than 1/16 inch from the diameter prior to the test.

4.12 **Coupling test.** A tensile force of not less than 25 pounds shall be applied to the engaged cleaner hose and tools for not less than 2 minutes and then the force released. Couplings shall conform to strength requirements of 3.15.6.1

4.13 **Inspection of preparation for delivery.** The preservation, packaging, packing and marking of the vacuum cleaners shall be examined to determine compliance with section 5 of this specification. Examination of packing requirements not covered by referenced specifications shall be in accordance with MIL-STD-105 using an AQL of 4.0 percent defective.

4.13.1 *Classification of defects (for packing and marking not covered by referenced specifications).*

101. Cushioning, blocking, bracing, and anchoring not as specified.
102. Case liner missing when required.
103. Case liner not sealed as specified.
104. Skids missing when required.
105. Closure of shipping container incorrect.
106. Gross weight of exterior shipping container exceeds limits specified.
107. Strapping missing when required.
108. Strapping not specified type and class or incorrectly applied.
109. Marking missing, illegible or incorrect.

5. PREPARATION FOR DELIVERY

(Civil agencies. Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of preservation, packaging and packing.)

5.1 **Preservation and packaging.** Preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 *Level A.* Unless otherwise specified, vacuum cleaners, attachments, accessories, and repair parts shall be preserved in accordance with MIL-P-116, method III. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1. When specified (see 6.1), complete vacuum cleaners shall be preserved in accordance with MIL-P-16298.

5.1.2 *Level C.* The complete vacuum cleaner shall be preserved and packaged in accordance with the industry's standard practice.

5.2 **Packing.**

5.2.1 *Level A.* One or more complete vacuum cleaners of identical type and class shall be packed in wood cleated-plywood boxes conforming to PPP-B-601, overseas type or nailed wood boxes conforming to

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PPP-B-621, class 2 or fiberboard box conforming to PPP-B-636, class 2, V3C or V3S, style RSC, when the weight of a vacuum cleaner does not exceed 1,000 pounds. The gross weight of boxes shall not exceed approximately 200 pounds, unless the weight of a vacuum cleaner exceeds this amount; fiberboard boxes shall be subject to the weight limitations of the fiberboard box specification. When the gross weight of a box exceeds 200 pounds, the box shall be modified by the addition of skids in accordance with the applicable box specification. When the weight of a complete vacuum cleaner exceeds 1,000 pounds, it shall be packed in a sheathed wood crate conforming to MIL-C-104, type I, style a. Contents of boxes and crates shall be cushioned, blocked, braced, and anchored in accordance with JAN-P-100 and the crate specification. Each box shall be provided with a waterproof case liner conforming to MIL-L-10547 and sealed in accordance with appendix thereto. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Flat-steel strapping shall be type I, class B.

5.2.2 *Level B.* One or more complete vacuum cleaners of identical type and class shall be packed in wood-cleated fiberboard, wirebound wood, wood cleated-plywood, nailed wood or fiberboard boxes conforming to PPP-B-591, domestic type; PPP-B-585, class 1; PPP-B-601, domestic type; PPP-B-621, class 1; and PPP-B-636, type I or II, class 1, style RSC, respectively, when the weight of a vacuum cleaner does not exceed 1,000 pounds. The gross weight of wood and wood-cleated boxes shall not exceed approximately 200 pounds, unless the weight of a vacuum cleaner exceeds this amount; fiberboard boxes shall be subject to the weight limitations of the fiberboard box specification. When the weight of a complete vacuum cleaner exceeds 1,000 pounds, the cleaner shall be packed in an open wood crate conforming to MIL-C-132, style B, class 1. Contents of boxes and crates shall be cushioned, blocked, braced, and anchored to prevent movement within containers.

5.2.3 *Level C.* Vacuum cleaners, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 *Civil agencies.* In addition to any special marking specified in the contract or order (see 6.1), shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, style and size desired (1.2.1).

(*c*) Specify if tanks of other than metal are desired (3.5.2).

(*d*) Specify "wet" pickup, if required (3.7.3).

(*e*) State voltage, current, whether a.c., d.c. or universal motors are required (3.9, 3.9.1 and table II).

(*f*) Specify explosion-proof thermal overload device, if required (3.9 and 3.11).

(*g*) Specify if radio filter is required (3.10).

(*h*) Whether radio-interference reduction is to conform to MIL-I-11748, MIL-I-26600 (USAF) or MIL-I-16910 (3.10.1).

(*i*) Specify grounding plug adapter, if required (see 3.12.2).

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(*j*) Specify requirements for attachments, if different (3.15, 3.16.8, 3.16.9, 3.17.6 and tables I and III; or if no attachments are required, so state).

(*k*) Specify all male couplings, if required (3.15.6).

(*l*) Specify if reversible machines are desired (3.16.1.1).

(*m*) State canister or cylinder model and separators, if one or the other is required (3.16.1.2 and 3.16.1.2.1).

(*n*) Specify type of mounting required (3.16.3).

(*o*) Specify if 2-inch wheels are not acceptable because of intended use (3.16.3.3).

(*p*) Specify handle adjustment desired if different from that specified (3.17).

(*q*) Specify if more than one cleaner of each type, class, size, and style shall be tested for radio-interference suppression (4.6.3).

(*r*) When vacuum cleaners are to be preserved in accordance with MIL-P-16298 (5.1.1).

(*s*) Levels of preservation, packaging and packing required (section 5).

(*t*) Marking, if different (5.3).

6.2 **Illustrations.** The illustrations shown herein are for the convenience of identification of styles only and are not intended to preclude the purchase of vacuum cleaners which are otherwise in accordance with the requirements of this specification.

6.3 **Intended use.** The vacuum cleaners covered by this specification are adaptable to the following and similar uses:

6.3.1 *Type I, classes A, B, and C cleaners.* Type I, classes A, B, and C cleaners (with separator receptacle capacities of 2,000, 690, and 360 cubic-inch minimum, respectively) are intended for 8-hour continuous operation. Cleaners with 3, 5, and 7-1/2 horsepower motors should be operated on electrical power circuits of 220-230 volts.

6.3.2 *Type I, class D cleaners.* Type I, class D cleaners are intended for cleaning a large number of small areas and for 8 hours' intermittent daily operation.

6.3.3 *Styles 1 and 2 cleaners.* Style 1 cleaners operate at lower speeds, are heavier, are equipped with longer hose and heavier and higher gage wire than style 2 cleaners. Motor sizes vary as shown in table I.

6.3.4 *Type I, class G cleaners.* Type I, class G cleaners are intended for use in hospitals or other areas where hygienic conditions must be maintained. The cleaners are intended for continuous operation for 8 hours daily.

6.3.5 *Type II, classes A and B cleaners.* Type II, classes A and B (size 1) vacuum cleaners are intended for cleaning a large number of small areas and for 8 hours' intermittent daily operation.

6.3.6 *Type II, class B (size 2).* Type II, class B, size 2 cleaners are intended for cleaning a large number of small areas and for 8 hours' daily operation.

6.3.7 *Type II, classes C and D cleaners.* Type II, classes C and D cleaners are intended primarily for upholstery and similar light cleaning.

6.3.8 *Type II, class F cleaners.* Type II, class F cleaners are of the exposed separator type with 1,200 cu. in. minimum receptacle capacity and high operating speeds. They are intended for continuous operation for 8 hours daily.

6.4 **Guarantee.** It is suggested that purchasing officer require a guarantee against defective material and workmanship for a period of one year from date of delivery and replacement made of all parts found defective within the guarantee period without cost to the Government.

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6.5 Instruction book. Purchasers should require that the vacuum cleaners offered in accordance with this specification be fully described and identified by type, class, size and style (where applicable). Descriptive literature for operation and use should be required.

6.6 Radio-interference reduction. The contracting officer should arrange for radio-interference reduction tests as specified in 4.6.3.

6.7 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.8 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Type I, class A, size 1, 2, 3, 4:

Rail:

Vacuum cleaners, mounted on wagons.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Vacuum cleaners, mounted on wagons.
Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

All Others:

Rail:

Vacuum cleaners, not otherwise indexed by name.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Vacuum cleaners, not otherwise indexed.
Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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